Algorithm Cheat Sheet — All Templates + Java Syntax

Conventions, Skeleton & Core API

Generic Test Harness

```
class Question {
    // ── PROBLEM-SOLVING PROCESS ─────────────────────────
    // 1. INPUT / OUTPUT   : types, sizes, ranges; what to return; examples.
    // 2. HIGH-LEVEL ALGO  : pick the pattern (two-ptr / BFS / DP / ...); state the idea in 1 line.
    // 3. IMPLEMENTATION   : data structures, loop/recursion, invariants.
    // 4. EDGE CASES       : empty, size 0/1, duplicates, negatives, overflow, all-same, not-found.
    // 5. COMPLEXITY       : time & space; is it within limits?
    // 6. OPTIMIZATION     : reduce time/space (hashing, two-pointer, rolling array, pruning, early exit).
    // ──────────────────────────────────────────────────

    private int data;
    public Question(int data) {
        this.data = data;
    }
    public void solve() {
        // TODO
    }
}
public class Solution {
    public static void main(String[] args) {
        Question q = new Question(0);
        q.solve();
    }
}
```

Naming Conventions (enforced across all template files)

Common Java API Cheat-Sheet

```
Map<Integer, Integer> map   = new HashMap<>();       // key → value
Map<Integer, List<Integer>> adj = new HashMap<>();   // adjacency / buckets
Set<Integer> set            = new HashSet<>();       // membership
List<Integer> list          = new ArrayList<>();     // dynamic array
Deque<Integer> stack = new ArrayDeque<>();           // STACK: push / pop / peek
Deque<Integer> queue = new ArrayDeque<>();           // QUEUE: offer / poll / peek

// (ArrayDeque beats java.util.Stack / LinkedList)
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
// min-heap (default)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]>   pq      = new PriorityQueue<>((a, b) -> a[1] - b[1]);
// by field, min-first
TreeMap<Integer, Integer> tmap = new TreeMap<>();    // sorted keys: floor/ceiling/first/last
TreeSet<Integer> tset          = new TreeSet<>();    // sorted set: floor/ceiling/higher/lower
LinkedHashSet<Integer> lset    = new LinkedHashSet<>(); // insertion-ordered set (LFU buckets)
Random rand                    = new Random();       // rand.nextInt(n) → 0..n-1 (RandomizedSet)
int[]       arr  = new int[n];
// defaults to 0
boolean[]   seen = new boolean[n];
// defaults to false
int[][]     grid = new int[m][n];
// 2D, all 0
int[]       dr   = {1,-1,0,0}, dc = {0,0,1,-1};  // DOWN,UP,RIGHT,LEFT; iterate for(dir=0..3) i+dr[dir],j+dc[dir]
```

```
// ── Map ──
map.put(k, v);
map.getOrDefault(k, 0);
// safe read with default
map.merge(k, 1, Integer::sum);
// ← counting frequency (most common)
map.computeIfAbsent(k, x -> new ArrayList<>()).add(v); // ← build adjacency / buckets
map.putIfAbsent(k, v);
// ← write only first occurrence (keep earliest index)
map.containsKey(k);  map.remove(k);
for (Map.Entry<Integer,Integer> e : map.entrySet()) { e.getKey(); e.getValue(); e.setValue(v); }
map.keySet();  map.values();
// ── Set ──
set.add(x);  set.contains(x);
set.remove(x);
// ── List ──
list.add(x);  list.get(i);  list.set(i, x);
list.remove(list.size() - 1);
// ← pop in backtracking
list.size();  list.isEmpty();
list.addAll(other);
Collections.sort(list);
// ascending
list.sort((a, b) -> b - a);
// custom / descending
Collections.reverse(list);
// ← reverse path after building it backwards
```

```
// Deque as STACK (LIFO)             // Deque as QUEUE (FIFO)
stack.push(x);
queue.offer(x);
int top = stack.pop();               int head = queue.poll();
int look = stack.peek();             int look = queue.peek();
stack.isEmpty();
queue.isEmpty();
// Deque BOTH ENDS — needed for the MONOTONE DEQUE (sliding-window max/min)
// and for addFirst-style level building (zigzag BFS).
deque.offerLast(x);    deque.pollLast();
deque.peekLast();      // back (push/pop/peek tail)
deque.offerFirst(x);   deque.pollFirst();
deque.peekFirst();     // front (push/pop/peek head)
deque.addLast(x);      deque.addFirst(x);
// same as offer*, throw if capacity-bound
// PriorityQueue (heap)
pq.offer(x);  int best = pq.poll();  int look = pq.peek();  pq.size();
pq.remove(x); // ← O(n) removal of an arbitrary element (lazy-deletion alternative in #480)
```

```
Integer.MAX_VALUE;   //  2147483647 (2^31 - 1)  — sentinel for "min so far"
Integer.MIN_VALUE;   // -2147483648 (-2^31)     — sentinel for "max so far"
Long.MAX_VALUE;      // when sums can overflow int, accumulate in long
int k = i + (j - i) / 2;                 // ← avoid (i + j) overflow in binary search
Integer.compare(a, b);                   // ← overflow-safe comparator; use instead of (a - b)
(a, b) -> Integer.compare(a[1], b[1]);// ← safe PriorityQueue/Arrays.sort comparator
Double.compare(a, b);                    // ← comparator for double[] heaps (probability, median)
```

```
ch - 'a';                      // 'a'..'z' → 0..25   (lowercase bucket index)
ch - '0';                      // '0'..'9' → 0..9    (digit value)
(char) ('a' + i);              // 0..25 → 'a'..'z'
num = num * 10 + (ch - '0');   // ← build a number digit by digit
Integer.parseInt("123");       // String → int (throws on bad input)
Integer.valueOf(123);          // int → Integer (boxed)
String.valueOf(123);           // int/char/etc → String
Character.getNumericValue('7');   // char → 7
// char classification
Character.isDigit(ch);
Character.isLetter(ch);
Character.isLetterOrDigit(ch);
Character.isWhitespace(ch);
Character.toLowerCase(ch);
Character.toUpperCase(ch);
```

```
s.length();  s.charAt(i);  s.substring(i, j);      // [i, j)
s.toCharArray();  s.split(" ");
s.equals(t);  s.compareTo(t);
s.indexOf("ab");  s.isEmpty();
s.startsWith("ab");  s.endsWith("z");
s.contains("x"); // prefix / suffix / substring tests
s.toLowerCase();  s.toUpperCase();
// case-normalize (e.g. case-insensitive compare)
s.trim();  s.replace('a', 'b');
s.repeat(n);          // strip ends / swap chars / repeat
StringBuilder builder = new StringBuilder();
builder.append(x);  builder.reverse();
builder.toString();
builder.charAt(i);  builder.setCharAt(i, c);  builder.deleteCharAt(i);
builder.length();
String.join(",", list);
// list → "a,b,c"
```

```
Arrays.sort(arr);
// ascending in place
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);      // 2D by first field
Arrays.fill(dp, Integer.MAX_VALUE);
// seed a dp/dist array
Arrays.equals(window, need);
// ← element-wise array compare (anagram check)
Arrays.copyOf(arr, len);
Arrays.copyOfRange(arr, i, j);
System.arraycopy(src, srcPos, dst, dstPos, len);    // ← fast block copy (merge step)
Arrays.asList(1, 2, 3);
// fixed-size List view
int sum = Arrays.stream(arr).sum();
int max = Arrays.stream(arr).max().getAsInt();
// List<Integer> ↔ int[]
int[] a = list.stream().mapToInt(Integer::intValue).toArray();
List<Integer> l = Arrays.stream(a).boxed().collect(Collectors.toList());
```

```
Math.max(a, b);  Math.min(a, b);
Math.abs(x);
Math.pow(a, b);  Math.sqrt(x);
Math.ceil(x);  Math.floor(x);
n & (n - 1);           // clear lowest set bit
n & (-n);              // isolate lowest set bit
1 << i;                // bit i set  (use 1L << i for i ≥ 31)
(n >> i) & 1;          // read bit i
(n & 1) == 0;          // ← even   (last bit 0)
(n & 1) == 1;          // ← odd    (last bit 1)
n >> 1;                // ← divide by 2 (drop last bit);  n << 1 = multiply by 2
Integer.bitCount(n);   // # of set bits
Integer.toBinaryString(n);
```

```
tmap.firstKey();  tmap.lastKey();
tmap.floorKey(x);     // largest key ≤ x
tmap.ceilingKey(x);   // smallest key ≥ x
tmap.lowerKey(x);     // largest key < x
tmap.higherKey(x);    // smallest key > x
tset.floor(x);  tset.ceiling(x);
tset.lower(x);  tset.higher(x);
tset.first();  tset.last();
```

```
// ── NEGATIVE-SAFE MODULO ── replaces the manual ((x % n) + n) % n
Math.floorMod(x, n);                 // always in [0, n-1] even when x < 0 (prefix-sum % k)
// ── int[] INSTEAD OF HashMap for a bounded key space ──
int[] count = new int[26];           // O(1) no-boxing freq vs HashMap<Character,Integer>
count[ch - 'a']++;                   // also faster to compare: Arrays.equals(a, b)
// ── 1D ROLLING ARRAY for DP ── drop a dimension when row i only needs row i-1
int[] dp = new int[n + 1];           // O(n) space instead of O(m·n)
for (...) for (int j = target; j >= w; j--) dp[j] += dp[j - w];  // 0/1 knapsack: iterate j DOWN
// ── FIXED-SIZE HEAP for top-k ── O(n log k) beats sorting O(n log n)
if (heap.size() > k) heap.poll();    // keep only k best; min-heap for "k largest"
// ── LAZY DELETION over pq.remove(obj) ── pq.remove is O(n); skip stale entries instead
if (entry[0] != dist[node]) continue; // Dijkstra: ignore outdated heap entries
// ── ArrayDeque over Stack / LinkedList ── faster, no synchronization, both ends O(1)
Deque<Integer> stack = new ArrayDeque<>();   // never use java.util.Stack
// ── StringBuilder over String += in a loop ── O(n) vs O(n²)
StringBuilder builder = new StringBuilder();  // append, then builder.toString() once
// ── EARLY EXIT / PRUNING ── return the moment the answer is decided
if (found) return result;            // e.g. backtracking bounds
```

Template Index

Two Pointers

Pattern 1 — Opposite Two Pointers

Variables:
i = left pointer · j = right pointer · scan range [i, j] closed

Pseudocode:

```
set i to the left end, j to the right end
while the pointers haven't crossed:
  if the current pair hits the target:
    record it, then step both ends inward
  else if the value is too small:
    move the left end right to grow it
  else (too large):
    move the right end left to shrink it
```

```
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        /* record */ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

Sliding Window

The Three Templates

Variables:
i = left edge (inclusive) · j = right edge / loop variable (inclusive), window = [i, j], size = j − i + 1 · n = array length · k = target window size · result = answer (max length, or min length seeded to MAX)

Pseudocode:

```
FIXED window of size k:
    i = 0
    for j = 0..n-1:
        add nums[j] to window state
        if window size (j - i + 1) == k:
            record the window (it is exactly size k)
            remove nums[i] from state; i = i + 1

MAX variable window (longest valid):
    i = 0; result = 0
    for j = 0..n-1:
        add nums[j] to state
        while window violates the constraint:
            remove nums[i] from state; i = i + 1
        result = max(result, j - i + 1)  (window is valid here)

MIN variable window (shortest valid):
    i = 0; result = +infinity
    for j = 0..n-1:
        add nums[j] to state
        while window is valid:
            result = min(result, j - i + 1)  (record while still valid)
            remove nums[i] from state; i = i + 1  (then shrink to try smaller)
```

```
int i = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    if (j - i + 1 == k) {
        // 2. record (window is exactly size k)
        // 3. remove nums[i] from state
        i++;
    }
}
int i = 0, result = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window exceeds constraint */) {
        i++;
    }
    result = Math.max(result, j - i + 1);
}
int i = 0, result = Integer.MAX_VALUE;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window VALID */) {
        result = Math.min(result, j - i + 1);
        i++;
    }
}
```

Frequency Map Trick (used by 567 and 76)

Variables:
need[c] = how many more of char c the window still needs (positive = still short, zero/negative = surplus) · required = total chars still missing; hits 0 when the window covers all of t · i = left edge, j = right edge

Pseudocode:

```
Expanding (add char at j):
    if need[char] was still positive (> 0): required = required - 1   (filled one more)
    need[char] = need[char] - 1

Shrinking (remove char at i):
    if need[char] was already 0 or surplus (>= 0): required = required + 1  (now short one)
    need[char] = need[char] + 1
    i = i + 1
```

```
if (need[s.charAt(j)]-- > 0) required--;
if (need[s.charAt(i)]++ >= 0) required++;
i++;
```

Prefix Sum + HashMap

Core Template

Variables:
map = prefixSum→count or index seen so far · sum = running prefix sum · lookup = the complement prefix we search for · i = current scan position

Pseudocode:

```
create empty map of prefix → (count or index)
seed map with prefix 0 (count=1 for counting, index=-1 for length)
sum = 0
for each index i:
    add nums[i] to sum
    lookup = transform(sum)          // sum-k, sum%k, etc.
    use map.get(lookup) to update result
    store sum (or sum%k) into map with count or index i
```

```
Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???);
long sum = 0;
for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum);
    map.put(storeKey(sum), storeValue(i)); // what to store (sum or sum%k → count or index)
}
```

Binary Search

The Only Template — [i, j), find first k with condition(k)

Variables:
[i, j) = half-open search range (answer lives in here) · k = midpoint · condition(k) = monotone test, false…false then TRUE…true; answer is the first k it holds

Pseudocode:

```
i = 0, j = length          # half-open [i, j)
while range non-empty (i < j):
    k = midpoint of i and j
    if condition(k) is false: answer is to the right, set i = k+1
    else: k might be the answer, set j = k
```

```
int i = 0, j = nums.length;
while (i < j) {
    int k = i + (j - i) / 2;
    if (!condition(k)) {
        i = k + 1;
    } else {
        j = k;
    }
}
```

How to Choose

Sorting

Template 1 — Merge Sort

Variables:
[start, end) half-open range being sorted · mid = split point · temp = scratch buffer · i = left scan pointer [start,mid) · j = right scan pointer [mid,end) · k = write pointer into temp

Pseudocode:

```
mergeSort(nums, start, end):
    if range has 0 or 1 elements: return // already sorted
    mid = midpoint of [start, end)
    mergeSort left half  [start, mid)
    mergeSort right half [mid, end)
    merge the two sorted halves

merge:
    i = start, j = mid, k = start
    while either half has elements left:
        pick smaller front of the two halves (left wins ties → stable)
        write it to temp[k], advance that half's pointer and k
    copy temp back into nums[start, end)
```

```
public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}
private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return;
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp);
    mergeSort(nums, mid,   end, temp);
    merge(nums, start, mid, end, temp);
}
private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

Variables:
[start, end) half-open range · pivot = chosen partition value · i = boundary of <pivot region · k = scan/cursor pointer · j = boundary of >pivot region (from the right)

Pseudocode:

```
quickSort(nums, start, end):
    if range has 0 or 1 elements: return
    pivot = middle element's value
    i = start, k = start, j = end-1     // [start,i)<p | [i,k)==p | [k,j] unknown | (j,end)>p
    while k <= j:
        if nums[k] < pivot:  swap into < region, advance k and i
        elif nums[k] == pivot: advance k
        else:                swap to > region, shrink j (don't advance k)
    recurse on < region [start, i)
    recurse on > region [k, end)         // ==pivot middle is final
```

```
public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    quickSort(nums, start, i);
    quickSort(nums, k, end);
}
private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

Template 3 — Quick Select (partial sort — k-th element)

Variables:
[start, end) half-open range · target = 0-indexed position we want settled · pivot = partition value · i = boundary of <pivot region · k = scan/cursor pointer · j = boundary of >pivot region

Pseudocode:

```
quickSelect(nums, start, end, target):
    pivot = middle element's value
    partition into [start,i)<p | [i,k)==p | [k,end)>p   // same 3-way as quick sort
    if target < i:       recurse into left [start, i)
    elif target < k:     target sits in ==pivot region → return pivot
    else:                recurse into right [k, end)
```

```
private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot;
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 — Counting Sort

Variables:
min/max = value range bounds · buckets[v−min] = count of value v · i = write pointer back into arr · v = current value index into buckets

Pseudocode:

```
if array empty: return
find min and max values
buckets = array sized (max-min+1), all zero
for each x in arr: increment buckets[x-min]
// tally occurrences
i = 0
for v from 0 to last bucket:
    while bucket v still has count: write (v+min) into arr[i], advance i
```

```
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

Linked List

When to Use — Signal → Pattern

All Four Templates

Variables:
sentinel = dummy node before head · previous = last kept node · current = node under inspection · next = saved successor · slow/fast = 1×/2× walkers · a/b = the two input lists

Pseudocode:

```
make sentinel pointing at head; previous = sentinel, current = head
while current exists:
    if current should be deleted: rewire previous.next to skip current
    else: advance previous to current
    advance current
return sentinel.next
previous = null, current = head
while current exists:
    save next = current.next
    point current backward at previous
    advance previous to current, current to next
return previous as new head
slow = head, fast = head
while fast and fast.next exist:
    advance slow one step, fast two steps
make sentinel; current = sentinel
while both a and b exist:
    splice the smaller head onto current; advance that list
    advance current
attach whichever list remains
return sentinel.next
```

```
// 1. DELETE / FILTER  - sentinel guards head removal; previous tracks last kept node
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;
    } else {
        previous = current;
    }
    current = current.next;
}
return sentinel.next;
// 2. REVERSAL - re-wire one node at a time
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;
    current.next = previous;
    previous = current;
    current = next;
}
return previous;
// 3. FAST / SLOW - two pointers at different speeds
ListNode slow = head, fast = head;
while (fast != null && fast.next != null)
{
    slow = slow.next;
    fast = fast.next.next;
}
// 4. MERGE - sentinel collects nodes from two lists in order
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;
```

String

Core Idioms

Variables:
count = frequency vector (index = char/digit bucket) · ch − 'a' = 0–25 bucket index · ch − '0' = 0–9 digit value · num = number built so far · builder = encoded/result string · buckets = lists indexed by frequency · expand(i,j) = palindrome length from a center

Pseudocode:

```
idiom 1 - char count for letters:
  make count of size 26
  for each char: count[ch-'a'] += 1
idiom 2 - char count for digits:
  make count of size 10
  for each char: count[ch-'0'] += 1
idiom 3 - char to integer:
  num = 0
  for each char left to right: num = num*10 + (char - '0')
idiom 4 - anagram hash key (frequency-based):
  count the 26 letters of s
  build a string from each letter label followed by its count
  return it (same key for all anagrams)
  alternative: sort the chars and use the sorted string as key
idiom 5 - bucket sort by frequency:
  count all ASCII chars
  make buckets indexed by frequency
  for each char with count>0: add it to buckets[its count]
  walk buckets from highest frequency down, append each char that many times
idiom 6 - expand from center:
  while i in bounds and j in bounds and chars at i and j match: move i left, j right
  return j - i - 1 (length of palindrome found)
```

```
// 1. CHAR COUNT - lowercase letters
int[] count = new int[26];
for (char ch : s.toCharArray()) count[ch - 'a']++;
// 2. CHAR COUNT - digits 0-9
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++;
// 3. CHAR TO INTEGER - build number digit by digit
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0');
// 4. ANAGRAM HASH KEY - frequency-based (bucket sort style) O(k) vs sort O(k log k)
String hashKey(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch - 'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i));
        builder.append(count[i]);
    }
    return builder.toString();
}
char[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars);
// 5. BUCKET SORT - sort by frequency without comparison sort
int[] count = new int[128];
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1]; // index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] = new ArrayList<>();
        buckets[freq].add((char) i);
    }
}
StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--) {
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++) builder.append(ch);
}
// 6. EXPAND FROM CENTER - palindrome check in O(1) space
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1;
}
```

Stack / Queue / Heap

All Templates

Variables:
stack = LIFO ArrayDeque (holds indices for monotone variants) · queue = monotone deque (front = window answer) · minPQ/maxPQ/pq = priority queue (heap) · maxHeap = lower half, minHeap = upper half for median

Pseudocode:

```
STACK: push/pop/peek all at the front (LIFO)
MONO-DECREASING (next greater): while current > stack top, pop and record current as its answer; push index
MONO-INCREASING (next smaller / span): while current < stack top, pop and compute width to new top boundary; push index
MONO-DEQUE (window max): drop back while smaller than current, push; drop front if out of window; front = window max
PRIORITY QUEUE: offer/poll/peek the min (or max with reverse comparator)
TWO HEAPS: push to maxHeap then shift its top to minHeap; rebalance so maxHeap >= minHeap; median from the tops
```

```
// 1. STACK (LIFO) - use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);
stack.pop();
stack.peek();
// 2A. MONOTONE DECREASING STACK - next GREATER element
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i];
    stack.push(i);
}
// 2B. MONOTONE INCREASING STACK - next SMALLER element / span width
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width  = stack.isEmpty() ? i : i - stack.peek() - 1;
    }
    stack.push(i);
}
// 3. MONOTONE DEQUE - sliding window max/min
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
        queue.pollLast();
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst();
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}
// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>();
PriorityQueue<Integer> maxPQ = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]>    pq    = new PriorityQueue<>((a, b) -> a[1] - b[1]);
// custom: sort by index 1
pq.offer(x);
pq.poll();
pq.peek();
// 5. TWO HEAPS - maintain median in O(log n)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll());
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll());
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}
```

BFS (Tree)

The Template — Level-Aware BFS

Variables:
queue = nodes waiting to be processed · size = level snapshot (count of nodes in the current level) · level = current depth counter · node = node polled this step · i = index within the level

Pseudocode:

```
make an empty queue
put root in the queue
level = 0
while the queue is not empty:
    size = how many nodes are in the queue right now (one whole level)
    level = level + 1
    repeat size times (index i = 0..size-1):
        node = take the front of the queue
        process node (level known; i==0 -> first, i==size-1 -> last)
        if node has a left child, add it to the queue
        if node has a right child, add it to the queue
```

```
Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        if (node.left  != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
}
```

When to Use — Signal → Pattern

The One Rule — Snapshot Level Size First

```
Always snapshot the level size BEFORE the inner for loop:
    int size = queue.size();
    for (int i = 0; i < size; i++) { ... }

Without the snapshot, newly added children mix with the current level
and i == size-1 / i == 0 checks become meaningless.
```

DFS / Backtracking

When to Use — Signal → Pattern

All Templates

Variables:
node = current tree node · accumulated/current = state carried down a path · answer/global = best cross-node result · grid[r][c] = cell at row r, col c · state[node] = 0 unseen / 1 in-stack / 2 done · start = first eligible index · current = in-progress candidate · nums[i] = element being chosen

Pseudocode:

```
if node null: return BASE; left=dfs(left); right=dfs(right); return combine(left,right,node.val)
if node null: return; accumulated+=node.val; if leaf: record; dfs(left,accumulated); dfs(right,accumulated)
if node null: return 0; left=max(0,dfs(left)); right=max(0,dfs(right)); answer=max(answer,left+right+node.val); return max(left,right)+node.val
if out of bounds or not TARGET: return; mark VISITED; recurse into 4 neighbors
if state==1: cycle; if state==2: safe; mark in-stack; recurse neighbors (cycle? return); mark done
if base: record copy; for i from start: skip dups; choose nums[i]; recurse(next); undo
```

```
// 1. PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)
private int dfs(TreeNode node) {
    if (node == null) return BASE;
    int left  = dfs(node.left);
    int right = dfs(node.right);
    return ...;
}
// 2. PROCESS NODE FIRST, PASS STATE DOWN (pre-order)
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left,  accumulated);
    dfs(node.right, accumulated);
}
// 3. TREE - GLOBAL ANSWER (return up the best single arm; update global across both arms)
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left  = Math.max(0, dfs(node.left));
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right + node.val);
    return Math.max(left, right) + node.val;
}
// 4. GRID DFS - mark visited by mutating grid; 4-directional flood fill
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}
// 5. GRAPH DFS - 3-color visited: 0=unseen 1=in-stack 2=done
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;
    if (state[node] == 2) return false;
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}
// 6. BACKTRACKING - choose / recurse / undo
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue;
        current.add(nums[i]);
        backtrack(next, current);
        current.remove(current.size()-1);
    }
}
```

Graph

Complexity Legend

```
BFS / DFS          O(V + E)              visit each vertex and edge once
Topo sort (Kahn)   O(V + E)              each node enqueued once, each edge relaxed once
Union-Find         ~O(n·α(n)) ≈ O(n)     α = inverse Ackermann, effectively constant
Dijkstra           O((V + E) log V)      non-negative weights ONLY
```

Graph Representation

```
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]);
}
List<List<int[]>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[]{edge[1], edge[2]});
int[] dr = {1, -1, 0,  0};
int[] dc = {0,  0, 1, -1};
```

Core Templates

Variables:
queue = BFS frontier · visited[] = seen flags · level = current distance ring · inDegree[] = remaining prerequisites per node · order = topo result · parent[]/rank[] = union-find forest · dist[] = shortest distance · pq = min-heap on cost · color[] = 2-coloring (-1/0/1)

Pseudocode:

```
BFS: mark start visited, enqueue; while queue, snapshot level size, pop each, push unvisited neighbors, increment level
TOPO: count inDegree per edge, enqueue all zero-inDegree, pop into order and decrement neighbors, enqueue new zeros
UNION-FIND: init parent[i]=i; find follows parent to root with path compression; union links roots by rank, false if same root
DIJKSTRA: dist[src]=0, push src; pop closest, skip if stale, relax each neighbor and push improved
BIPARTITE: color each component start 0, BFS painting neighbors opposite, return false on same-color clash
```

```
// 1. BFS - shortest path / level order (track distance by level-size snapshot)
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    for (int i = 0; i < size; i++) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.offer(neighbor);
            }
        }
    }
    level++;
}
// 2. TOPOLOGICAL SORT - Kahn's BFS
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0) queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0) queue.offer(neighbor);
}
// 3. UNION FIND
int[] parent, rank;
void init(int n) {
    parent = new int[n]; rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}
int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]);
    return parent[x];
}
boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}
// 4. DIJKSTRA - shortest path (non-negative weights)
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[src] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
pq.offer(new int[]{src, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], d = current[1];
    if (d > dist[node]) continue;
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next])
        {
            dist[next] = dist[node] + w;
            pq.offer(new int[]{next, dist[next]});
        }
    }
}
// 5. BIPARTITE CHECK - BFS 2-coloring
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] != -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) { color[neighbor] = 1 - color[node]; queue.offer(neighbor); }
            else if (color[neighbor] == color[node]) return false;
        }
    }
}
return true;
```

Greedy

Two Interval Templates

```
MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):
    Sort by START  →  traverse FORWARD  (i = 0 → n-1), extend the END of last kept
    Sort by END    →  traverse BACKWARD (i = n-1 → 0), extend the START of last kept
    These two are exact mirror images. See #56 for both written out side by side.
    (NOTE: this mirror only applies to merging. Template 1 below sorts by end but
     still sweeps FORWARD - there the goal is counting, not merging.)

TEMPLATE 1 - Sort by END time, sweep forward, track end of last kept interval
            Use when: maximizing count of non-overlapping intervals
            Greedy insight: earliest-finishing interval leaves most room for future ones

TEMPLATE 2 - Sort by START time, sweep forward, merge when overlapping
            Use when: merging/counting overlapping intervals
            Greedy insight: processing in start order → each interval only needs to
            compare with the last merged result
```

Variables:
intervals = list of [start, end] · lastEnd = end of last kept interval · result = merged output · pq = min-heap of active end times

Pseudocode:

```
TEMPLATE 1 (sort by end, count non-overlapping):
  sort intervals by END time
  lastEnd = -infinity
  for each interval:
    if interval.start >= lastEnd:   # no overlap
      lastEnd = interval.end
      keep it (count++ or add)
    else:
      skip it (overlap)

TEMPLATE 2 (sort by start, merge):
  sort intervals by START time
  for each interval:
    if result empty OR last kept end < interval.start:
      add interval as new
    else:
      extend last kept end = max(last end, interval.end)

TEMPLATE 3 (sort by start + min-heap of ends):
  sort intervals by START time
  for each interval:
    if heap nonempty AND earliest end <= interval.start:
      pop heap              # reuse freed resource
    push interval.end       # occupy a resource
  heap size = resources needed
```

```
Arrays.sort(intervals, (a, b) -> a[1] - b[1]);
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) {
        lastEnd = interval[1];
    } else {
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval);
    } else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1], interval[1]);
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0])
        pq.poll();
    pq.offer(interval[1]);
}
```

Dynamic Programming

All Six Templates

Variables:
dp[i] = the answer for the subproblem at index/state i (meaning varies per family: ways/cost/length/feasibility ending at or reaching i) · n = problem size · target/j = knapsack capacity dimension.

Pseudocode:

```
LINEAR 1D:    dp[i] = fixed recipe over dp[i-1], dp[i-2] (Kadane: best ending at i = max(start fresh, extend))
LOOK-BACK:    for each i, scan all earlier j and extend the best valid dp[j]
KNAPSACK:     collapse item dimension to 1D; descending j = each item once, ascending j = reusable
2D SEQUENCE: match consumes both strings (diagonal), mismatch drops one side
INTERVAL:     solve short ranges first, combine via a split point inside the range
GRID:         each cell accumulates from cells it could arrive from (up / left)
STATE MACHINE: name a few states, update each from yesterday's states every step
```

```
// 1. LINEAR 1D - each cell depends on a fixed number of previous cells
int[] dp = new int[n + 1];
dp[0] = base;
for (int i = 1; i <= n; i++)
    dp[i] = f(dp[i-1], dp[i-2], ...);
// KADANE (max subarray, #53) = Linear-1D where dp[i] = best sum ENDING at i:
// 2A. LOOK-BACK 1D - dp[i] depends on all j < i
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]);
    }
}
// 3. 2D SEQUENCE - match consumes both, mismatch drops one side. WHEN: LCS / edit
// 4. INTERVAL DP - short ranges first (i right-to-left, j from i+1; dp[i+1][*] ready).
// 5. GRID DP - each cell accumulates from cells it could arrive from (up/left).
// 6. STATE MACHINE - named states updated from yesterday's states each step.
```

```
Want count or min/max?
├── Count (dp[j] += dp[j - num])
│   ├── 0/1 (each item once):       j descending → #416, #494
│   └── Unbounded (reuse):          j ascending → #518
│       └── Ordered (permuts):      target outer, nums inner → #377
└── Min/Max (dp[j] = min/max(...))
    ├── 0/1 minimize:               j descending → #474 (maximize)
    └── Unbounded minimize:         j ascending → #322
```

Variables:
dp[j] = number of ways to reach sum j (count flavor; dp[0]=1 is the empty selection); i = item index, j = current capacity/target, num = an item's weight/value. Loop *direction* decides reuse: descending j = item used at most once, ascending j = item reusable; target-outer = orderings counted.

Pseudocode:

```
0/1 KNAPSACK (each item once):
    dp[0] = 1
    for each item i:
        for j from target DOWN TO nums[i]:
// DESCENDING reads OLD dp (item i not yet used)
            dp[j] += dp[j - nums[i]]

UNBOUNDED KNAPSACK (item reusable):
    dp[0] = 1
    for each item i:
        for j from nums[i] UP TO target:
// ASCENDING reads NEW dp (item i already used this pass)
            dp[j] += dp[j - nums[i]]

PERMUTATION COUNT (order matters):
    dp[0] = 1
    for j from 1 to target:
// OUTER = target value
        for each num in nums:
// INNER = try every num, so orderings differ
            if j >= num: dp[j] += dp[j - num]
```

```
// ── 0/1 KNAPSACK ── each item at most once
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = target; j >= nums[i]; j--)
        dp[j] += dp[j - nums[i]];
}
// ── UNBOUNDED KNAPSACK ── each item reusable
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = nums[i]; j <= target; j++)
        dp[j] += dp[j - nums[i]];
}
// ── PERMUTATION COUNT ── order matters (Combination Sum IV)
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {
    for (int num : nums) {
        if (j >= num) dp[j] += dp[j - num];
    }
}
```

Variables:
dp[i][j] = the answer for the prefixes s1[0..i) (first i chars) and s2[0..j) (first j chars); index 0 = empty prefix.

Pseudocode:

```
dp = (m+1) x (n+1) table
initialize dp[0][*] and dp[*][0] from the empty-prefix base cases
for i from 1 to m:
// each char of s1
    for j from 1 to n:
// each char of s2
        if s1[i-1] == s2[j-1]:
// current chars match
            dp[i][j] = dp[i-1][j-1] + 1
// extend the diagonal (both prefixes shrink by 1)
        else:
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])  // drop a char from one/both side
return dp[m][n]
```

```
int[][] dp = new int[m + 1][n + 1];
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1;
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
```

Variables:
dp[i][j] = the answer for the interval [i, j] (left endpoint i, right endpoint j).

Pseudocode:

```
dp = n x n table
for each i: dp[i][i] = base_case
// length-1 intervals
for i from n-1 down to 0:
// i RIGHT-TO-LEFT so dp[i+1][*] (inner) is ready
    for j from i+1 to n-1:
// j LEFT-TO-RIGHT from i+1 so dp[i][j-1] is ready
        // safe to read dp[i+1][j], dp[i][j-1], dp[i+1][j-1] (all strictly smaller intervals)
        dp[i][j] = ...transition over interval [i,j]...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = n - 1; i >= 0; i--) {
    for (int j = i + 1; j < n; j++) {
        dp[i][j] = ...;
    }
}
```

Variables:
dp[i][j] = the answer for the interval [j, i] (right endpoint i, left endpoint j).

Pseudocode:

```
dp = n x n table
for each i: dp[i][i] = base_case
// length-1 intervals
for i from 0 to n-1:
// i = RIGHT endpoint, FORWARD left-to-right
    for j from i-1 down to 0:
// j = LEFT endpoint, DESCENDING from i-1 to 0
        // safe to read dp[i-1][j], dp[i][j+1], dp[i-1][j+1] (all shorter intervals already filled)
        dp[i][j] = ...transition over interval [j,i]...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = 0; i < n; i++) {
    for (int j = i - 1; j >= 0; j--) {
        dp[i][j] = ...;
    }
}
```

Variables:
dp[i][j] = the count/answer for the (i, j) subproblem, defined only for j <= i (lower-triangular).

Pseudocode:

```
dp = n x n table
for each i: dp[i][0] = 1
// column base case; dp[0][j>0] = 0
for i from 1 to n-1:
// fill ROW BY ROW, top to bottom
    for j from 1 to i:
// j from 1 up to i (triangular, j <= i)
        // safe to read dp[i-1][j], dp[i][j-1], dp[i-1][j-1] (prior row / earlier in row)
        dp[i][j] = ...transition...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][0] = 1;
for (int i = 1; i < n; i++) {
    for (int j = 1; j <= i; j++) {
        dp[i][j] = ...;
    }
}
```

Variables:
dp[i][j] = best (path count / cost) to reach cell (i, j) moving only right/down; memo[i][j] = longest path starting from (i, j) for the all-direction DFS variant; dr/dc = 4-direction deltas.

Pseudocode:

```
dr = {1,-1,0,0}; dc = {0,0,1,-1}
// DOWN UP RIGHT LEFT

// Standard grid (only right/down - DAG, no cycle):
dp = rows x cols table
initialize first row and first column
for i from 0 to rows-1:
    for j from 0 to cols-1:
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1])   // combine top & left

// All-direction grid (memoized DFS for increasing/decreasing paths):
memo = rows x cols table
for i from 0 to rows-1:
    for j from 0 to cols-1:
        dfs(grid, i, j, memo, dr, dc)
// each cell caches its longest path
```

```
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1};
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dfs(grid, i, j, memo, dr, dc);
```

```
cash     = max profit when NOT holding (free to buy or idle)
hold     = max profit when HOLDING stock (bought it at some cost)
cooldown = max profit right after selling (can't buy today)
```

```
                        buy                                  sell
re-buy condition
#121 (1 tx):            hold = max(hold, -price)             no re-buy (ignore cash)
#122 (unlimited):       hold = max(hold, cash - price)       re-buy with full cash
#123 (2 tx):            chain: buy2 uses sell1 cash          4 states chained
#309 (cooldown):        hold = max(hold, cooldown - price)   must wait 1 day
#714 (fee):             hold = max(hold, cash - price)       pay fee on sell
```

Bit Manipulation

Canonical Template

Variables:
result = XOR accumulator (lone survivor) · n = number being inspected/edited · i = bit position · count = set-bit counter · mask = 26-bit letter-presence set · masks: 1 << i selects bit i, n & (n−1) clears lowest set bit, n & (−n) isolates lowest set bit

Pseudocode:

```
XOR CANCEL:
    result = 0
    for each num in nums: result = result XOR num   (pairs cancel, lone survives)

BIT OPERATIONS on n at position i:
    isSet  = bit i of n is 1            -> (n >> i) & 1 == 1
    set    = turn bit i on              -> n | (1 << i)
    clear  = turn bit i off             -> n & ~(1 << i)
    toggle = flip bit i                 -> n ^ (1 << i)
    lowest = isolate lowest set bit     -> n & (-n)
    isPow2 = n > 0 and clearing lowest set bit gives 0

COUNT SET BITS (Brian Kernighan):
    count = 0
    while n != 0: count = count + 1; n = n & (n-1)   (each step removes one set bit)

BITMASK AS SET:
    mask = 0
    for each char c in word: mask = mask | (1 << (c - 'a'))
    two words share no letter when (mask[i] & mask[j]) == 0
```

```
// ── XOR CANCEL ── pairs cancel; lone element survives
int result = 0;
for (int num : nums) result ^= num;
// ── BIT OPERATIONS ──
boolean isSet  = ((n >> i) & 1) == 1;
int set        = n | (1 << i);
int clear      = n & ~(1 << i);
int toggle     = n ^ (1 << i);
int lowest     = n & (-n);
boolean isPow2 = n > 0 && (n & (n-1)) == 0;
// ── COUNT SET BITS ── Brian Kernighan: each iteration removes one set bit
int count = 0;
while (n != 0) { count++; n &= (n - 1); }
// ── BITMASK AS SET ── 26-bit integer represents presence of 26 letters
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

Variations

Variables:
ones/twos = bits seen 1 / 2 times mod 3 · xor = a ^ b of the two uniques · diff = lowest differing bit (xor & −xor) · a/b = the two unique values · dp[] = set-bit counts by number · shift = bits dropped to reach common prefix · carry = AND-derived carry bits

Pseudocode:

```
VARIATION 1 (mod-3 state machine, Single Number II):
    ones = 0; twos = 0
    for each num: ones = (ones XOR num) AND NOT twos; twos = (twos XOR num) AND NOT ones
    after loop, ones holds the element appearing once

VARIATION 2 (split XOR into two groups, Single Number III):
    xor = XOR of all nums                    (= a XOR b)
    diff = xor AND (-xor)  (lowest bit where a and b differ)
    a = 0; for each num: if (num AND diff) != 0: a = a XOR num   (XOR one group)
    b = xor XOR a  (the other unique)

VARIATION 3 (DP relation, Counting Bits):
    dp = array of size n+1
    for i = 1..n: dp[i] = dp[i >> 1] + (i AND 1)   (drop last bit, add it back)

VARIATION 4 (common prefix, Bitwise AND of Range):
    shift = 0
    while left != right: left >>= 1; right >>= 1; shift = shift + 1
    return left << shift  (shared high-bit prefix restored)

VARIATION 5 (carry loop, Sum of Two Integers):
    while b != 0:
        carry = a AND b  (positions that carry)
        a = a XOR b  (sum without carry)
        b = carry << 1  (carry moves one bit left)
    return a
```

```
int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;
    twos = (twos ^ num) & ~ones;
}
int xor  = 0;
for (int num : nums) xor ^= num;
int diff = xor & (-xor);
int a = 0;
for (int num : nums)
    if ((num & diff) != 0) a ^= num;
int b = xor ^ a;
int[] dp = new int[n + 1];
for (int i = 1; i <= n; i++)
    dp[i] = dp[i >> 1] + (i & 1);
int shift = 0;
while (left != right) { left >>= 1; right >>= 1; shift++; }
return left << shift;
int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;
        a = a ^ b;
        b = carry << 1;
    }
    return a;
}
```

Design

Canonical Template — HashMap + Doubly Linked List (LRU)

Variables:
map = key -> CacheNode · head/tail = sentinels (head side = MRU, tail side = LRU) · node.previous/node.next = doubly linked neighbors

Pseudocode:

```
CacheNode: key, value, previous, next
init: head <-> tail (head.next = tail; tail.previous = head)

remove(node):
  node.previous.next = node.next
  node.next.previous = node.previous

insertAfterHead(node):          # mark as most-recently-used
  link node between head and head.next

evict LRU:
  lru = tail.previous           # node just before tail
  remove(lru); map.remove(lru.key)
```

```
class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}
// ── SENTINEL NODES ── head (MRU side) ↔ ... ↔ tail (LRU side)
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
head.next = tail; tail.previous = head;
// ── REMOVE NODE ──
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}
// ── INSERT AFTER HEAD (= mark as MRU) ──
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}
// ── LRU EVICTION ── tail.previous is LRU
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);
```

Variations

Variables:
keyToVal/keyToFreq = key lookups · freqToKeys = freq -> ordered keys at that freq · minFreq = smallest live frequency · (RandomizedSet) list = values array · map = val -> index

Pseudocode:

```
VARIATION 1 (LFU): maps key->value, key->freq, freq->LinkedHashSet<key>, plus minFreq
  on access: freq++; move key from freqToKeys[old] to freqToKeys[new]
  on evict: remove first key (oldest) from freqToKeys[minFreq]

VARIATION 2 (RandomizedSet): ArrayList of values + HashMap val->index
  remove: swap target with last element, update map, drop last
  getRandom: list[random index]
```

Math

Canonical Templates

Variables:
n = the number being processed · digit = last digit peeled · x = base in fast power · result = running product/accumulator · b = the base for conversion · composite[] = sieve marks

Pseudocode:

```
DIGIT EXTRACTION:
  while n is not zero:
    digit = remainder of n divided by 10
    drop the last digit of n
    use digit

FAST POWER (x^n):
  result starts at 1
  while exponent n still has bits:
    if the lowest bit is set, fold the current x into result
    square x to reach the next power
    shift n right to consume that bit
  return result

BASE CONVERSION (number -> digits):
  while num is positive:
    append num mod b
    divide num by b
  reverse the collected digits
BASE CONVERSION (digits -> number):
  value starts at 0
  for each digit: value = value * b + digit

SIEVE:
  make a composite[] flag array up to n
  for i from 2 while i*i < n:
    if i already marked composite, skip it
    mark every multiple of i starting at i*i as composite
```

```
// ── DIGIT EXTRACTION LOOP ── peel digits right-to-left
while (n != 0) {
    int digit = n % 10;
    n /= 10;
}
// ── FAST POWER (exponentiation by squaring) ──
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x;
        x *= x;
        n >>= 1;
    }
    return result;
}
// ── BASE CONVERSION ── generic base b, 0-indexed
StringBuilder builder = new StringBuilder();
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
builder.reverse();
int value = 0;
for (int digit : digits) value = value * b + digit;
// ── SIEVE OF ERATOSTHENES ── mark composites up to n
boolean[] composite = new boolean[n];
for (int i = 2; i * i < n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j < n; j += i)
        composite[j] = true;
}
```